

# Midhun Rajan

## Manufacturing & Process Engineer | Manufacturing Systems | Digital Transformation & AI Automation

Luton, Bedfordshire, UK | midhunrajan74@gmail.com | +44 7445 588889 | github.com/MR4693 | linkedin.com/in/midhun-rajan

### PROFESSIONAL SUMMARY

Manufacturing and process engineer with experience across sheet metal design, fabrication and ERP-driven production in India, the UAE and the UK since 2017. Today I work inside a live precision sheet metal manufacturing environment, with a dual focus on production engineering and the software systems that run the floor. Over 2025 I led a digital transformation programme that moved core operations from paper to a substantially digitised, ERP-integrated production environment — specifying, designing, validating and deploying four systems — built using AI-assisted software development under my own engineering direction — now in active daily use across 19 operators and 14 work centres.

The work spans the full path from operational problem to deployed system: shop floor digitisation, real-time production intelligence, and the automation of engineering document and job pack workflows — all integrated against MIE Trak ERP on factory infrastructure. Every requirement was identified through direct operational involvement, and I continue to maintain and improve each system in daily production use.

### EXPERIENCE

|                                  |                 |
|----------------------------------|-----------------|
| Manufacturing & Process Engineer | 2025 - Present  |
| Technical Engineer               | Nov 2022 - 2025 |

#### Ludwick Precision Ltd · Precision Sheet Metal Manufacturing · Hitchin, Hertfordshire, UK

Promoted to Manufacturing & Process Engineer in 2025 after joining as Technical Engineer in November 2022.

- ✓ Manufacturing engineering across a multi-work-centre shop floor — CNC fibre laser cutting, tube laser processing, CNC press brake, welding, fabrication and assembly — including sheet metal modelling, DFM review and ERP-driven production planning.
- ✓ Led an end-to-end digital transformation programme, identifying operational problems from daily floor involvement and delivering four ERP-integrated systems rather than purchasing commercial software — using AI-assisted software development under my own engineering direction.
- ✓ Designed and deployed the **Manufacturing Operations Platform** — Android shop floor terminals for clock-on, digital job packs, live work orders, move tickets and sign-off — eliminating paper documentation and end-of-shift ERP data entry across 19 operators.
- ✓ Designed and delivered the **Manufacturing Intelligence Platform** — real-time management dashboards over live MIE Trak data — giving directors and supervisors continuous shop floor visibility without ERP access or floor walks.
- ✓ Designed and delivered the **Digital Job Pack Platform** — Excel-driven ERP lookup with automated PDF extraction and assembly — reducing job pack preparation from hours to seconds per batch.
- ✓ Designed and delivered the **Document Intelligence Platform** — drag-and-drop intake with automatic part number detection and automated MIE Trak ERP filing for PDF, STEP and DXF files — replacing manual document control with a system-enforced process.
- ✓ Defined the requirements, designed the workflows and business logic, validated and tested every feature, and deployed each system into live manufacturing operations — and continue to maintain and improve all four on factory infrastructure, with no external development resource or third-party support contracts.

## Design Engineer

Apr 2021 – Jun 2022

### Dolphin Manufacturing LLC · Sheet Metal Manufacturing · Ajman, UAE

- ✓ Designed sheet metal components and assemblies for manufacture using 3D CAD, optimised for laser cutting, bending and fabrication.
- ✓ Produced manufacturing drawings, DXF files and supporting documentation for the shop floor.
- ✓ Worked closely with production teams to ensure designs were practical, manufacturable and cost-effective.
- ✓ Supported product development, design improvements and engineering change implementation.

## Fabrication Engineer

Jul 2017 – Aug 2020

### R.S. Franklin Associates · Structural & Sheet Metal Fabrication · Nagercoil, Tamil Nadu, India

- ✓ Prepared fabrication and manufacturing drawings for structural and sheet metal projects.
- ✓ Planned fabrication processes and coordinated production activities to ensure quality and on-time delivery.
- ✓ Supported shop-floor operations, material planning and quality inspections.
- ✓ Worked with production teams to resolve manufacturing issues and improve fabrication efficiency.

## SELECTED ACHIEVEMENTS

### PAPERLESS SHOP FLOOR

19 operators and 14 work centres moved onto a live digital platform — paper job packs, clock cards and move tickets replaced with a traceable, ERP-linked workflow.

### REAL-TIME ERP DATA

Production records now update the moment operators act — eliminating end-of-shift manual data entry and the data lag that came with it.

### HOURS TO SECONDS

Job pack preparation automated from a 30-minute-to-several-hour manual task to seconds per batch, against live ERP document records.

### MANAGEMENT VISIBILITY

Four-view intelligence platform delivering continuous, real-time production visibility to directors and supervisors with no specialist ERP knowledge required.

## EDUCATION

### Bachelor of Engineering (Honours), Mechanical Engineering

Anna University · Chennai, Tamil Nadu, India

### Master Diploma in Product Design

CADD Centre · Trivandrum, Kerala, India

### Professional Diploma in HVAC & MEP Design

CADD Centre · Trivandrum, Kerala, India

## SKILLS & CAPABILITIES

Positioned by what I deliver. My core expertise is manufacturing engineering and digital process design; the software was built using AI-assisted development under my own direction. The technologies under Technical Exposure are tools I have practical working knowledge of and used to deliver these systems — not presented as specialist software-engineering expertise.

### MANUFACTURING SYSTEMS · CORE EXPERTISE

Manufacturing Process Engineering

ERP Integration (MIE Trak)

SQL Data Integration

Manufacturing Workflow Design

Digital Process Automation

Manufacturing Intelligence

Technical Solution Architecture

### AI & DIGITAL TECHNOLOGIES

AI-assisted Software Development

Process Automation

Dashboard Development

Manufacturing Analytics

Document Intelligence

System Integration

### TECHNICAL EXPOSURE · WORKING KNOWLEDGE

Technologies I have practical, working knowledge of — used to build and maintain the systems above with AI-assisted development. Not presented as expert-level programming.

- Python
- SQL
- Flask
- HTML
- CSS
- JavaScript

MANUFACTURING DISCIPLINES

- CNC Fibre Laser Cutting
- Tube Laser Processing
- CNC Press Brake
- Welding & Fabrication
- Assembly
- SolidWorks
- Sheet Metal DFM
- Laser Nesting
- Production Planning
- ERP-Driven Manufacturing

ENGINEERING APPROACH

**ON AI-ASSISTED ENGINEERING**

The systems in this portfolio were implemented using AI-assisted engineering, with an AI model used as a software development partner under full engineer